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Discrete Mathematics 236 (2001) 179–189

DISCRETE
MATHEMATICS

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On a solution of a randomized three tracks variant of the Gate Matrix Layout problem[☆]

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Received 2 December 1997; revised 15 July 1998; accepted 5 March 2000

Abstract

We are concerned with the k -Gate Matrix Layout (k -GML) problem in a very large scale integration design. The problem asks if k tracks are enough to lay out a given circuit. The k -GML problem may be stated equivalently in terms of random Boolean matrices. In the paper, we find thresholds for the appearance of matrices which are ‘yes’ instances for a randomized 3-GML problem. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: k -Gate Matrix Layout; Boolean matrix; Random Boolean matrix; Forbidden intersection graph; Forbidden bipartite generator; Subgraph threshold

1. The k -GML problem

We are concerned here with a combinatorial optimization problem known in the literature as the *Gate Matrix Layout (GML)* problem. GML is a simple layout style for very large scale integration (VLSI) circuits (see [7]) and it uses the structure of Boolean matrix. The circuit components are the vertical lines which are polysilicon lines serving as transistor gates and/or conductors (‘gates’), and the horizontal tracks which are groups of transistor diffusion or metal lines (‘nets’). There is a vertical gate line for each discrete input, and a transistor with that input must be placed over that gate. Additionally, all transistors associated in the same net must be placed in a common horizontal track.

Various algorithms have been developed [1–4] to automate the layout procedure. The final goal is to reduce the total chip area by reducing the number of horizontal tracks required to lay out the circuit. There may be more than one net in the same

[☆] Research supported in part by grant KBN 2P03A 023 09.

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track, but only if the nets do not overlap (i.e., two nets can be in the same tracks if they are different and one of them is incident on all of the required gate lines before the second begins).

We construct a gate Boolean matrix R by creating a column for each gate and a row for each net. A 1 in position (i, j) represents a transistor that is in net i and must be placed on gate j . Our goal is to find a column permutation of R so that the nets of 1's can be packed densely in the smallest number of rows (two nets can be in the same row if the rightmost 1 of one of them lies on the left of the leftmost 1 of the second).

The GML problem is known in general to be NP-hard (see [5,6,8,12]). Consider a related decision problem known in the literature as the k -GML. An $n \times m$ Boolean matrix R and an integer k are given. We ask whether we can permute the columns of R so that, if in each row we change to * every 0 lying between the row's leftmost and rightmost 1, then no column contains more than k 1's and *'s. Such a * is called a *fill-in*. We say that a given Boolean matrix R is a 'yes' instance for k -GML if such a permutation exists.

Let matrix $R = R(n, m, p)$ be a random Boolean matrix. That is, the entries of R are 0's and 1's; each entry equals either 1 with probability p or 0 with probability $1 - p$, independently of each other. Throughout the paper, we always assume that $m = n^\alpha$, where $\alpha > 0$ is a chosen real number. Note that we have $nm p$ 1's on average in R and we may now ask the question: what is the probability that a random Boolean matrix is a 'yes' instance for k -GML.

Our main result, given below, states for which range of p and α , the probability that a 'large' (i.e., as $n \rightarrow \infty$) Boolean matrix is a 'yes' instance for 3-GML tends to one, as well as when this probability tends to zero.

Theorem 1. *Let $R(n, m, p)$ be a random Boolean matrix, $m = n^\alpha$, where $\alpha > 0$ and*

$$p_3^* = p_3^*(n) = \begin{cases} m^{-1/4} n^{-1} & \text{for } \alpha \leq \frac{40}{21}, \\ m^{-1/2} n^{-11/21} & \text{for } \alpha \geq \frac{40}{21}. \end{cases}$$

Then

$$\lim_{n \rightarrow \infty} \Pr(R(n, m, p) \text{ is a 'yes' instance for 3-GML}) = \begin{cases} 1 & \text{if } p \ll p_3^*, \\ 0 & \text{if } p \gg p_3^*. \end{cases}$$

2. Graph representation of the k -GML problem

Let R be an $n \times m$ Boolean matrix. With each such matrix we can associate an *intersection graph* G in the following way.

Let $V(G) = \{1, 2, \dots, n\} = [n]$ and let $E(G)$ be the set of edges of G defined as follows: for every $u, v \in V(G)$, there is an edge $\{u, v\} \in E(G)$ if and only if the row u and the row v of matrix R have 1's in at least one column (i.e., the dot product of rows u and v is nonzero). Any matrix R which generates a given graph G is called

its *incidence matrix*. Note that there are many incidence matrices associated with the same intersection graph G . The family of all incidence matrices of the graph G is denoted by $\mathcal{R}_G$.

We say that a graph G is a ‘yes’ instance for k -GML if for at least one matrix $R \in \mathcal{R}_G$, k -tracks is enough to layout R . Otherwise, we say G is a ‘no’ instance for k -GML.

The incidence matrix R can be viewed also as the adjacency matrix of a bipartite graph $B = (V_1, V_2, E(B))$, where $V_1 = [n]$, $V_2 = [m]$ and there is an edge $\{u, z\} \in E(B)$ if and only if there is 1 in the crossing of u th row and z th column of R . For convenience, we shall call vertices from V_2 the ‘*top*’ vertices of B , while vertices from V_1 will be called ‘*bottom*’ vertices.

Similarly, one can generate an intersection graph G from B . Now, $V(G) = [n] = V_1$ (‘bottom’ vertices of B), and there is an edge $\{u, v\} \in E(G)$ if and only if there exists a vertex $z \in V_2$ (a ‘top’ vertex of B), such that $\{u, z\}, \{v, z\} \in E(B)$. There are many bipartite graphs B generating the same graph G . We denote a set of those bipartite graphs $\mathcal{B}_G$ and note that there is obvious one-to-one correspondence between $\mathcal{B}_G$ and $\mathcal{R}_G$.

Recall that a graph G is a *minor* of the graph H , denoted $G \leq_m H$, if and only if any graph isomorphic to G can be obtained from H by a series of the following two operations: taking subgraph and contracting an arbitrary edge. (To find more information about minors see [15,16].) We say that a graph G is an *obstruction* for k -GML if it is a ‘no’ instance for k -GML and it is minimal in the minor order. Denote by $\mathcal{F}_k^{\leq_m}$ the set of all graphs which are obstructions for k -GML and call such set the *obstruction set*.

For 1-GML, it is trivial to see that the obstruction set consists just of K_2 . For 2-GML the only obstructions are K_3 and $S(K_{1,3})$, where $S(K_{1,3})$ is a graph obtained by subdividing each edge of the star $K_{1,3}$. Kinnersley and Langston (see [14]) constructed the set of obstructions for 3-GML, i.e., a set of minor-minimal graphs that represent a ‘no’ instance for 3-GML and proved the following result.

Theorem 2 (Kinnersley and Langston [14]). *There are exactly 110 graphs which are obstructions for 3-GML.*

In the paper [14], a complete list of those 110 graphs is given. The reader is encouraged to consult that paper since we shall restrict ourselves to a general description of that obstruction set only.

One can notice that all obstructions for 3-GML are planar graphs. The smallest element of $\mathcal{F}_3^{\leq_m}$, in terms of its number of vertices, is a complete graph on four vertices (K_4) and the largest are ten trees on 22 vertices. There are 23 graphs with exactly one interior face, 39 with two, 29 with three and just 9 graphs with exactly four interior faces, respectively. Interior faces in all those graphs are cycles of length at least 3 and at most 9.

For $k \geq 4$, it seems to be a very difficult task to identify and construct all members of the obstruction set for k -GML. For example, the obstruction set for $k=4$ contains at

least 122 million elements (see [13]). However, one may easily notice that a complete graph K_{k+1} is always an obstruction for k -GML. We also conjecture that, for any k , trees on $t_k = (5 \times 3^{k-1} - 1)/2$ vertices are the largest elements of the obstruction set (then $t_1=2$, $t_{k+1}=3 \times t_k + 1$, so that for $k=2, 3, 4, 5$ and 6 values of t_k are 7, 22, 67, 202 and 607, respectively). We shall return to that problem in the conclusion of our paper.

Note that if a graph G is an obstruction for k -GML, i.e., $G \in \mathcal{F}_k^{\leq m}$, then a proper minor of G is a ‘yes’ instance for k -GML. However G and every graph H such that $G \leq_m H$ are ‘no’ instances for k -GML. Denote by $\mathcal{F}_k$ the set of all graphs that are ‘no’ instances for k -GML and call elements of $\mathcal{F}_k$ *forbidden intersection graphs* for k -GML.

Recall that $\mathcal{B}_H$ denotes the set of all bipartite graphs which generate an intersection graph H . Hence $\mathcal{B}_k = \bigcup_{H \in \mathcal{F}_k} \mathcal{B}_H$ can be analogously called the set of *forbidden bipartite generators* for k -GML.

3. Randomized 3-GML

In Section 1, we stated our main result in the language of random Boolean matrices. In light of the discussion from the previous section, one can immediately see that the problem can be reformulated and stated in a random graph theory language (see [9]). Indeed, finding the probability that a random Boolean matrix $R(n, m, p)$ is a ‘no’ instance for 3-GML is equivalent to establishing the probability that a *random intersection graph* $G(n, m, p)$ contains a subgraph which itself, or its minor, is forbidden for 3-GML.

By a *random intersection graph* we mean a graph generated by $R(n, m, p)$ in the way described in Section 2. In fact, one can give an equivalent definition of such a random graph which more precisely explains why we call such a graph an intersection graph. Namely, we say that graph $G(m, n, p)$ is a random intersection graph if for any n, m positive integers, so that $m = n^\alpha$, $p \in [0, 1]$, and for every positive integer u with $1 \leq u \leq n$, we can assign S_u a random subset of $[m] = \{1, 2, \dots, m\}$ formed by selecting each element of $[m]$ independently with probability p . Then $\{u, v\} \in E(G)$ exactly when $S_u \cap S_v \neq \emptyset$. One can easily notice that the presence of edges in a random intersection graph $G(n, m, p)$ can depend on absence or presence of other edges, so we no longer deal with a model which assumes edge independence (see [11, 17]).

Hence, instead of proving our main result (Theorem 1) in algebraic terms we shall prove its graph theoretical analog stated below.

Theorem 3. *Let $G(n, m, p)$, where $m = n^\alpha$, be a random intersection graph. If H is a forbidden intersection graph for 3-GML, (i.e., $H \in \mathcal{F}_3$) and*

$$p_3^* = p_3^*(n) = \begin{cases} m^{-1/4} n^{-1} & \text{for } \alpha \leq \frac{40}{21}, \\ m^{-1/2} n^{-11/21} & \text{for } \alpha \geq \frac{40}{21}, \end{cases}$$

then

$$\lim_{n \rightarrow \infty} \Pr(H \subseteq G(n, m, p)) = \begin{cases} 0 & \text{if } p \ll p_3^*, \\ 1 & \text{if } p \gg p_3^*. \end{cases}$$

Since edges in a random intersection graph are not independent, establishing directly the probability that $H \subseteq G(n, m, p)$ is a rather difficult task. However, we can instead look at a random bipartite graph $B(n, m, p)$ whose adjacency matrix $R(n, m, p)$ generates $G(n, m, p)$ and then look for those subgraphs of $B(n, m, p)$ which are forbidden generators of graphs from $\mathcal{F}_3$, i.e., for subgraphs belonging to the family $\mathcal{B}_3$. Then, to prove Theorem 3, and also Theorem 1, we can use an earlier result of Karoński and Ruciński [10] given below.

Let us first define two notions we need to state this result. Consider a subgraph B of a complete bipartite graph with bipartition $([n], [m])$, $m = n^\alpha$ and assume that B has e_B edges, x_B ‘bottom’ vertices and y_B ‘top’ vertices. Then

$$d_\alpha(B) = \frac{e_B}{\alpha y_B + x_B}$$

is called α -density of B . Moreover,

$$m_\alpha(B) = \max_{L \subseteq B} d_\alpha(L)$$

is called *maximum subgraph α -density* of the graph B .

Lemma 4 (Karoński and Ruciński [10]). *Let $B(n, m, p)$, where $m = n^\alpha$, be a random bipartite graph. If B is a bipartite graph with x_B ‘bottom’ vertices, y_B ‘top’ vertices and e_B edges, and*

$$p^* = p_B^* = n^{-1/m_\alpha(B)},$$

then

$$\lim_{n \rightarrow \infty} \Pr(B \subseteq B(n, m, p)) = \begin{cases} 0 & \text{if } p \ll p^*, \\ 1 & \text{if } p \gg p^*. \end{cases}$$

Since the above threshold can be established for any forbidden generator $B \in \mathcal{B}_3$, i.e., for any element of forbidden generators of all graphs which are ‘no’ instances for 3-GML, to complete the proof one has to find those subgraphs from $\mathcal{B}_3$ with the ‘earliest appearance’ in a random graph $B(n, m, p)$. Notice that it depends solely on the α -density of such graphs.

To proceed further, we should make one more observation. Namely, if we have two bipartite graphs, B_1 and B_2 , such that $B_1 \subseteq B_2$, then $m_\alpha(B_1) \leq m_\alpha(B_2)$. To see this note that

$$\begin{aligned} m_\alpha(B_2) &= \max_{L \subseteq B_2} d_\alpha(L) = \max \left\{ \max_{L \subseteq B_2, L \not\subseteq B_1} d_\alpha(L), \max_{L \subseteq B_1} d_\alpha(L) \right\} \\ &= \max \left\{ \max_{L \subseteq B_2, L \not\subseteq B_1} d_\alpha(L), m_\alpha(B_1) \right\} \geq m_\alpha(B_1). \end{aligned}$$

Hence to establish our main result, we have to consider only those bipartite generators which are minimal in the subgraph order. For further considerations we assume that $\mathcal{B}_H$ no longer denotes the set of all bipartite generators of a given graph H , but the set of *minimal bipartite generators* of H , and consequently, $\mathcal{B}_3$ is the set of minimal bipartite generators of all graphs which are ‘no’ instances for 3-GML.

Let

$$m_\alpha(\mathcal{B}_H) = \min_{B \in \mathcal{B}_H} m_\alpha(B),$$

and let

$$m_\alpha(\mathcal{B}_3) = \min\{m_\alpha(\mathcal{B}_{K_4}), m_\alpha(\mathcal{B}_{T_{22}})\}.$$

The second crucial element of the proof of our main result is the following lemma, which we shall prove in the next section.

Lemma 5. *Let $\mathcal{F}_3$ be the family of forbidden intersection graphs for 3-GML. Then*

$$\min_{H \in \mathcal{F}_3} m_\alpha(\mathcal{B}_H) = m_\alpha(\mathcal{B}_3),$$

where

$$m_\alpha(\mathcal{B}_3) = \begin{cases} 4/(\alpha + 4) & \text{for } \alpha \leq \frac{40}{21}, \\ 42/(21\alpha + 22) & \text{for } \alpha \geq \frac{40}{21}. \end{cases}$$

Now we are ready to complete the proof.

Proof of Theorem 3. Note that by Lemma 4, it is enough to show that

$$p_3^* = \min_{H \in \mathcal{F}_3} p_H = \min_{H \in \mathcal{F}_3} \min_{B \in \mathcal{B}_H} p_B^*.$$

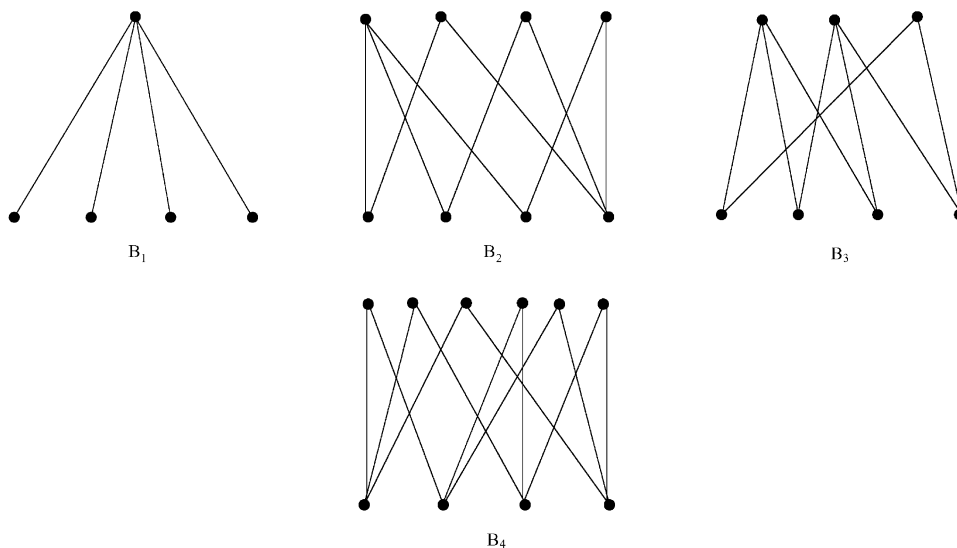
However, by Lemma 5 we have

$$\begin{aligned} p_3^* &= \min_{H \in \mathcal{F}_3} \min_{B \in \mathcal{B}_H} n^{-1/m_\alpha(B)} = \min_{H \in \mathcal{F}_3} n^{-1/\min_{B \in \mathcal{B}_H} m_\alpha(B)} \\ &= \min_{H \in \mathcal{F}_3} n^{-1/m_\alpha(\mathcal{B}_H)} = n^{-1/\min_{H \in \mathcal{F}_3} m_\alpha(\mathcal{B}_H)} \\ &= n^{-1/m_\alpha(\mathcal{B}_3)} \end{aligned}$$

and the theorem follows. $\square$

4. Special obstructions for 3-GML

There are eleven special graphs which are obstructions for 3-GML: K_4 and ten trees on twenty two vertices. Given one of these 10 trees on 22 vertices, we denote it by T_{22} .

Fig. 1. Minimal generators of K_4 .

Lemma 6.

$$m_\alpha(\mathcal{B}_{K_4}) = \begin{cases} 4/(\alpha + 4) & \text{for } \alpha \leq \frac{8}{3}, \\ 6/(3\alpha + 2) & \text{for } \alpha \geq \frac{8}{3}. \end{cases}$$

Proof. It is easy to show that the set $\mathcal{B}_{K_4}$ of minimal generators of K_4 contains only four bipartite graphs. (Fig. 1)

Hence, $\mathcal{B}_{K_4} = \{B_1, B_2, B_3, B_4\}$ and $m_\alpha(\mathcal{B}_{K_4}) = \min_{B \in \mathcal{B}_{K_4}} m_\alpha(B) = \min_{B \in \mathcal{B}_{K_4}} \max_{L \subseteq B} d_\alpha(L)$.

Notice that

$$m_\alpha(B_1) = \frac{4}{\alpha + 4},$$

$$m_\alpha(B_2) \geq d_\alpha(B_2) = \frac{9}{4\alpha + 4},$$

$$m_\alpha(B_3) \geq d_\alpha(B_3) = \frac{8}{3\alpha + 4},$$

$$m_\alpha(B_4) = \frac{12}{6\alpha + 4} = \frac{6}{3\alpha + 2}.$$

Therefore,

$$m_\alpha(\mathcal{B}_{K_4}) = \min_{B \in \mathcal{B}_{K_4}} m_\alpha(B) = \begin{cases} m_\alpha(B_1) & \text{for } \alpha \leq \frac{8}{3}, \\ m_\alpha(B_4) & \text{for } \alpha \geq \frac{8}{3}. \end{cases} \quad \square$$

Lemma 7. For every $\alpha > 0$,

$$m_\alpha(\mathcal{B}_{T_{22}}) \geq \frac{42}{21\alpha + 22},$$

with equality holding for all $\alpha \geq \frac{20}{21}$.

Proof. Since the same argument holds for any tree, we shall prove in fact a more general result. Namely, let T_t denote a tree on t vertices. Such a tree can be generated in an intersection graph by a single minimal bipartite generator, namely by a tree on $2t - 1$ vertices such that t of the vertices are ‘bottom’ vertices, and the $t - 1$ remaining vertices are the ‘top’ vertices. Denote such a tree by B . Then

$$m_\alpha(\mathcal{B}_{T_t}) = m_\alpha(B) = \max_{L \subseteq B} d_\alpha(L).$$

Let $B_{i,j}$ denote a subgraph of B formed by deleting i bottom vertices with i edges adjacent to those vertices and j top vertices with j edges adjacent to those vertices. Of course $B_{0,0} = B$ and so

$$d_\alpha(B) = \frac{2t - 2}{\alpha(t - 1) + t}.$$

Also,

$$d_\alpha(B_{i,j}) = \frac{2t - 2 - i - j}{\alpha(t - 1 - j) + t - i}.$$

It is easily observed that for $\alpha \geq 1 - 1/(t - 1)$,

$$\frac{2t - 2}{\alpha(t - 1) + t} \geq \frac{2t - 2 - i - j}{\alpha(t - 1 - j) + t - i}.$$

Hence for such α ,

$$m_\alpha(\mathcal{B}_{T_t}) = \max_{i,j} d_\alpha(B_{i,j}) = d_\alpha(B)$$

and obviously $m_\alpha(\mathcal{B}_{T_t})$ can be bounded from below by $d_\alpha(B)$, for any positive α .

The thesis of Lemma 7 follows directly from the above statement. $\square$

From the two last lemmas, one can deduce the following fact.

Corollary 8. Let $m_\alpha(\mathcal{B}_3) = \min\{m_\alpha(\mathcal{B}_{K_4}), m_\alpha(\mathcal{B}_{T_{22}})\}$. Then

$$m_\alpha(\mathcal{B}_3) = \begin{cases} 4/(\alpha + 4) & \text{for } \alpha \leq \frac{40}{21}, \\ 42/(21\alpha + 22) & \text{for } \alpha \geq \frac{40}{21}. \end{cases}$$

Proof of Lemma 5 from Section 3. From the above corollary it follows that Lemma 5 simply states that the complete graph K_4 and the trees T_{22} dominate the entire obstruction set consisting of 110 graphs. This follows since the lemma claims that

$$m_\alpha(\mathcal{B}_3) = \min_{H \in \mathcal{F}_3} m_\alpha(\mathcal{B}_H).$$

It holds if we are able to show that indeed for every $H \in \mathcal{F}_3$ and for every $B \in \mathcal{B}_H$,

$$m_\alpha(\mathcal{B}_3) \leq m_\alpha(B). \quad (1)$$

To see this take any graph $H \in \mathcal{F}_3$ and consider all minimal bipartite generators of H having minimum ‘top’ degree at least four. Then such a generator induces in H a subgraph isomorphic to K_4 and the statement (1) easily follows.

Hence, one has to consider only those bipartite generators whose ‘top’ vertices have degrees equal to 2 or 3. Notice that the vertices of degree one can be excluded since we deal with minimal generators only.

Let $y_{B,3}$ and $y_{B,2}$ denote the number of ‘top’ vertices of degree three and two, respectively. Moreover, let $c_{B,3}$ denote the number of connected components of a subgraph of H induced by the ‘top’ vertices of B of degree three only.

Recall that y_B and x_B denote the number of ‘top’ and ‘bottom’ vertices of the bipartite generator B , while e_B denotes its number of edges. Here $y_B = y_{B,3} + y_{B,2}$. Furthermore, let v_H and e_H stand for the number of vertices and edges of the graph H generated by B , respectively. Then the following relationships are true:

$$v_H \leq y_{B,3} + y_{B,2} + c_{B,3} + 1,$$

$$e_H = 2y_{B,3} + y_{B,2} + c_{B,3}.$$

Moreover,

$$x_B = v_H, \quad y_B = e_H - y_{B,3} - c_{B,3},$$

while

$$e_B = 2e_H - y_{B,3} - 2c_{B,3}.$$

Therefore,

$$d_\alpha(B) = \frac{e_B}{y_B\alpha + x_B} = \frac{2e_H - y_{B,3} - 2c_{B,3}}{(e_H - y_{B,3} - c_{B,3})\alpha + v_H}$$

and to prove (1), notice that for $\alpha \leq \frac{40}{21}$,

$$\frac{4}{\alpha + 4} \leq \frac{2e_H - y_{B,3} - 2c_{B,3}}{(e_H - y_{B,3} - c_{B,3})\alpha + v_H},$$

while, for $\alpha \geq \frac{40}{21}$,

$$\frac{42}{21\alpha + 22} \leq \frac{2e_H - y_{B,3} - 2c_{B,3}}{(e_H - y_{B,3} - c_{B,3})\alpha + v_H}.$$

One can easily check that the two preceding inequalities, for such range of α , can be reduced to the inequality

$$22e_H - 21v_H \geq 22c_{B,3} - 9y_{B,3}. \quad (2)$$

Since $c_{B,3} \leq y_{B,3}$, let us assume first that $c_{B,3} = y_{B,3} \geq 1$. Now,

$$22e_H - 21v_H = 21(e_H - v_H) + e_H \geq 21(y_{B,3} - 1) + 2y_{B,3} + y_{B,2} + c_{B,3}.$$

Hence (2) holds if $24y_{B,3} - 21 \geq 13y_{B,3}$, which is true for all $y_{B,3} \geq 2$.

If $y_{B,3} = c_{B,3} = 1$ then inequality (2) reduces to

$$21(e_H - v_H) + e_H \geq 13,$$

which is trivially true when $e_H > v_H$.

In the case $e_H = v_H$, we have to check whether $v_H \geq 13$. Then H is a connected graph with the same number of vertices and edges, so H is a graph with exactly one cycle having trees attached to it. Here the cycle is a triangle generated in H by a single ‘top’ vertex of degree three. However, $H \in \mathcal{F}_3$ and so there is a graph G in the set $\mathcal{F}_3^{\leq m}$ of 110 obstructions for 3-GML, such that $G \leq_m H$. Therefore, G must be either a triangle with trees attached to it or a tree itself. One can check (see the list given in [14]) that all such graphs from $\mathcal{F}_3^{\leq m}$ have at least 13 vertices.

Notice that when B has ‘top’ vertices of degree two only, i.e., when $y_{B,3} = c_{B,3} = 0$, then inequality (2) reduces to the trivially satisfied inequality $21(e_H - v_H) + e_H \geq 0$.

To complete the proof we have to consider the case when vertices of degree three have some common neighborhoods, i.e., when $c_{B,3} < y_{B,3}$. Then

$$21(e_H - v_H) + e_H \geq 21(y_{B,3} - 1) + 2y_{B,3} + y_{B,2} + c_{B,3} \geq 22c_{B,3}$$

and the lemma follows. $\square$

5. Conclusions

In this paper, we have established the threshold for the appearance of random Boolean matrices that are ‘no’ instances for 3-GML. Although it seems to be very hard to find a set of forbidden graphs for k -GML for $k \geq 4$, we know however that K_{k+1} is the smallest element of the set of obstructions and we think that the trees on $t_k = (5 \times 3^{k-1} - 1)/2$ vertices are the largest elements of this set. Therefore, we conjecture that the following statement holds.

Conjecture. Let $R(n, m, p)$ be a random matrix, $m = n^\alpha$, where $\alpha \geq 0$ and

$$p_k^* = p_k^*(n) = \begin{cases} m^{-1/(k+1)} n^{-1} & \text{for } \alpha \leq \frac{(t_k-2)(k+1)}{(t_k-1)(k-1)}, \\ m^{-1/2} n^{-t_k/2(t_k-1)} & \text{for } \alpha \geq \frac{(t_k-2)(k+1)}{(t_k-1)(k-1)}. \end{cases}$$

Then

$$\lim_{n \rightarrow \infty} \Pr(R(n, m, p) \text{ is a ‘yes’ instance for } k\text{-GML}) = \begin{cases} 1 & \text{if } p \ll p_k^*, \\ 0 & \text{if } p \gg p_k^*. \end{cases}$$

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